Bike Sharing System Analysis

Attribute (Column /Features) Details:

Attribute Name	*Data Type*	*Description*
Station ID	*General*	*Unique identifier assigned to each bike station*
Station Number	*Number*	*Numeric code used to identify the bike station within the system*
Station Name	*Text*	*Name of the bike station*
Address	*Text*	*Physical location or street address of the bike station*
Banking	*Text*	*Indicates whether the station supports bank card payment (Yes/No).*
Bonus	*Text*	*Shows if the station offers bonus rewards for returning bikes.*
Status	*Text*	*Current operational status of the station (Open/Closed).*
City	*Text*	*City where the bike station is located.*
Latitude	*General*	*Geographic coordinate representing the north-south position of the station.*
Longitude	*General*	*Geographic coordinate representing the east-west position of the station.*
Bike stands	*Number*	*Total number of bike parking stands available at the station.*
Available Bike stands	*Number*	*Number of empty stands available for returning bikes.*
Available Bikes	*Number*	*Number of bikes currently available for users to rent.*
Last Updated Date	*Date*	*Date when the station information was last updated.*
Last Updated Time	*Time*	*Time when the station information was last updated.*
Time zone	*Time*	*Time zone of the station location.*

Tools & Technologies

Excel: *Data cleaning, transformation, Sorting, Created separate tables using filters, Pivot Tables and Chart*

Power BI: *Data modeling, DAX measures & Visualization*

Data Pre-Processing (Excel)

Tasks Performed:

Data Cleaning & Transformation:

- *Renamed column headers and corrected data types for consistency.*
- *Created a new* ***Station ID column*** *by combining* Station Number *and* City *using the* ***CONCAT function****.*
- *Separated the* ***Position field into Latitude and Longitude****.*
- *Replaced incorrect or inconsistent values in* ***Station Name and Address*** *columns.*
- *Replaced* ***TRUE/FALSE values with Yes/No*** *in the* ***Banking and Bonus columns*** *to improve readability and consistency in the dataset.*
- *Split the* ***Last Updated column*** *into* ***Date, Time, and Time Zone*** *using* ***Text to Columns (Fixed Width)****.*
- *Performed* ***data transformation in Power Query****.*
- *Applied* ***Trim and Clean*** *functions to remove extra spaces and unwanted characters.*
- *Standardized text formatting by* ***capitalizing each word*** *in* Banking, Bonus, Status, and City *columns.*
- *Split the* ***Address column using a delimiter*** *to separate location details.*

Filtering & Sorting:

- *Applied filters to identify* ***missing values and data inconsistencies****.*
- *Sorted the data to organize records and verify data accuracy.*

Pivot Tables:

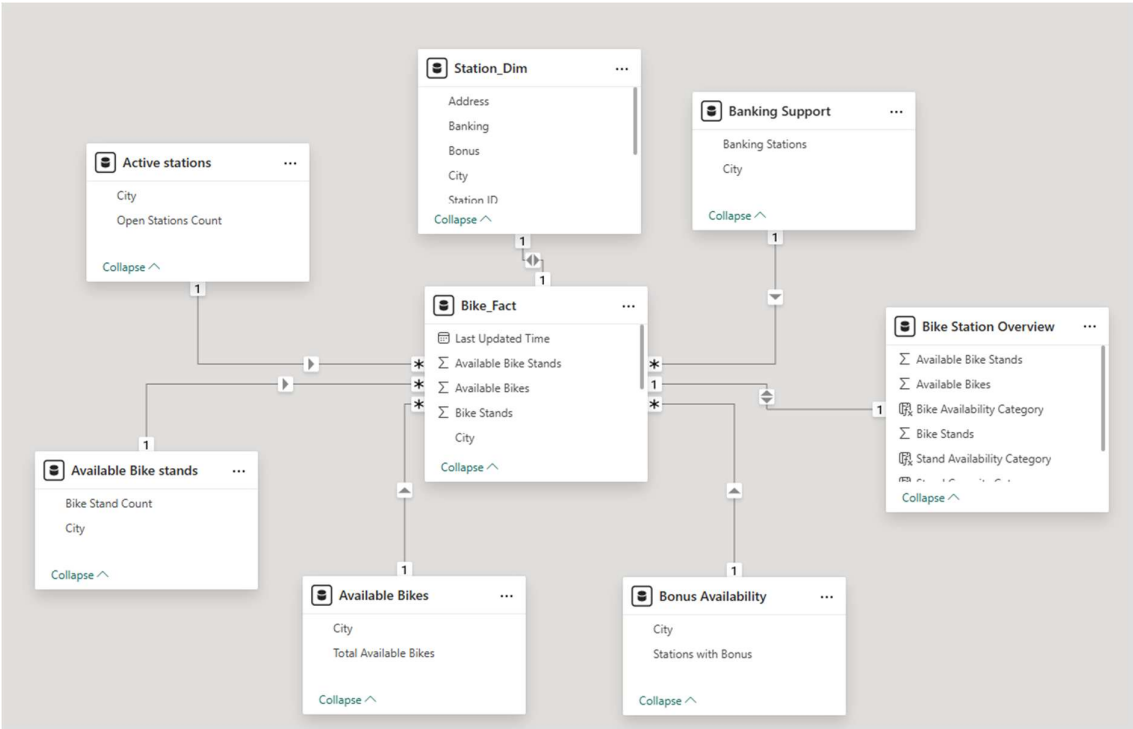
- *Created a* ***pivot table to analyze the annual count of available bikes and bike stands****.*
- *Developed another* ***pivot table to identify the distribution of bike stations across different cities****.*
- *Created a* ***relationship between tables using the primary key (Station ID)*** *to ensure proper data connection and accurate analysis.*

Chart:

- ***Created a*** *line chart* ***to visually analyze the*** *annual count of available bikes and bike stands****, supported by*** *timeline slicers for interactive time-based analysis****.***
- ***Developed a*** *column chart* ***to analyze the*** *distribution of bike stations across cities****, with*** *slicers for Status, Bonus, and Banking* ***to enable dynamic filtering.***

Data Modelling and DAX (Power BI)

Data Model: *Created a* ***data model consisting of 7 tables*** *to organize the dataset and establish relationships for efficient analysis and reporting.*



Calculated Columns & DAX Measures:

Caculated table: DAX formula

- ***Active stations =***

```
SUMMARIZE(FILTER(Station_Dim,Station_Dim[Status]="Open"),Station_Dim[City ],"Open Stations Count",CALCULATE(COUNTROWS(Station_Dim)))
```

City	Open Stations Count
Amiens	67
Lillestrom	3
Santander	6
Creteil	10
Lyon	440
Cergy-Pontoise	14
Rouen	10
Mulhouse	16
Lund	22
Toyama	25
Ljubljana	86
Seville	271
Valence	277
Luxembourg	147
Namur	30
Toulouse	429
Nantes	128
Bruxelles-Capitale	348
Maribor	40
Nancy	40
Dublin	116
Besancon	34
Vilnius	1
Jcdecauxbike	12
Marseille	78

- ***Available Bike stands*** = SUMMARIZE(filter(Bike_Fact,Bike_Fact[Available Bike Stands]>0),Bike_Fact[City],"Bike Stand Count", CALCULATE(sum(Bike_Fact[Available Bike Stands])))

City	Bike Stand Count
Amiens	726
Lillestrom	116
Santander	212
Creteil	136
Lyon	5014
Cergy-Pontoise	435
Rouen	195
Mulhouse	504
Lund	169
Toyama	196
Ljubljana	951
Seville	3133
Valence	3119
Luxembourg	1715
Namur	311
Toulouse	4720
Nantes	1588
Bruxelles-Capitale	4435
Maribor	456
Nancy	416
Dublin	1325
Besancon	360
Vilnius	346
Marseille	1450
Jcdecauxbike	90

- ***Available Bikes*** = SUMMARIZE(FILTER(Bike_Fact,Bike_Fact[Available Bikes]>0),Bike_Fact[City],"Total Available Bikes",CALCULATE(SUM(Bike_Fact[Available Bikes])))

City	Total Available Bikes
Amiens	366
Lyon	3654
Cergy-Pontoise	450
Mulhouse	75
Ljubljana	500
Seville	1873
Valence	2116
Luxembourg	1043
Toulouse	3583
Nantes	782
Bruxelles-Capitale	2569
Nancy	182
Dublin	693
Jcdecauxbike	89
Marseille	1020
Creteil	1
Rouen	29
Lund	83
Toyama	32
Namur	154
Maribor	143
Besancon	55
Vilnius	117
Lillestrom	15
Santander	22

- ***Banking Support*** =
  SUMMARIZE(filter(Station_Dim,Station_Dim[Banking]="Yes"),Station_Dim[City],"Banking Stations",CALCULATE(COUNTROWS(Station_Dim)))

City	Banking Stations
Amiens	3
Santander	18
Creteil	10
Lyon	20
Cergy-Pontoise	1
Rouen	28
Mulhouse	9
Lund	5
Valence	1
Namur	1
Toulouse	62
Nantes	33
Nancy	7
Besancon	7
Jcdecauxbike	2
Marseille	129

- ***Bonus Availability*** =
  SUMMARIZE(FILTER(Station_Dim,Station_Dim[Bonus]="Yes"),Station_Dim[City],"Stations with Bonus",CALCULATE(COUNTROWS(Station_Dim)))

City	Stations with Bonus
Lyon	2
Bruxelles-Capitale	58
Besancon	1
Marseille	9

- ***Bike Station Overview*** = SUMMARIZE(Bike_Fact,Bike_Fact[Station ID],Bike_Fact[Bike Stands],Bike_Fact[Available Bike Stands],Bike_Fact[Available Bikes])

Station ID	Bike Stands	Available Bike Stands	Available Bikes
1_Amiens	1	0	0
1_Besancon	17	0	0
1_Bruxelles-Capitale	17	0	0
1_Cergy-Pontoise	10	0	0
1_Creteil	15	15	0
1_Dublin	14	0	0
1_Lillestrom	9	0	0
1_Ljubljana	10	10	0
1_Lund	10	0	0
1_Luxembourg	10	0	0
1_Lyon	0	0	0
1_Maribor	1	0	1
1_Mulhouse	20	13	7
1_Namur	8	0	0
1_Nancy	10	0	0
1_Nantes	16	0	0

Caculated Column: DAX formula

Stand Capacity Category = SWITCH(TRUE(),
'Bike Station Overview'[Bike Stands]=0,"No Capacity",

'Bike Station Overview'[Bike Stands]>0 && 'Bike Station Overview'[Bike Stands]<=20,"Small",

'Bike Station Overview'[Bike Stands]>20 && 'Bike Station Overview'[Bike Stands]<=30,"Medium",

'Bike Station Overview'[Bike Stands]>30 && 'Bike Station Overview'[Bike Stands]<=50, "Large",

"Very Large")

Stand Capacity Category
Small
Small
Small
Small
Small
Small
Small
Small
Small
Small
No Capacity
Small
Small
Small
Small
Small
Medium

Stand Availability Category = SWITCH(true(),
'Bike Station Overview'[Available Bike Stands]=0,"Full",

'Bike Station Overview'[Available Bike Stands]>0 && 'Bike Station Overview'[Available Bike Stands]<=20,"Low",

'Bike Station Overview'[Available Bike Stands]>20 && 'Bike Station Overview'[Available Bike Stands]<=35, "Medium",

"High")

Stand Availability Category
Full
Full
Full
Full
Low
Full
Full
Low
Full
Full
Full
Full
Low
Full
Full
Full
Medium

Bike Availability Category = SWITCH(TRUE(),
'Bike Station Overview'[Available Bikes]=0,"Not Available",

'Bike Station Overview'[Available Bikes]>0 && 'Bike Station Overview'[Available Bikes]<=20,"Low",

'Bike Station Overview'[Available Bikes]>20 && 'Bike Station Overview'[Available Bikes]<=35,"Medium",

"High")

Bike Availability Category
Not Available
Not Available
Not Available
Not Available
Not Available
Not Available
Not Available
Not Available
Not Available
Not Available
Not Available
Low
Low
Not Available
Not Available
Not Available
Not Available

Calculated Measures: DAX formula

- ***Banking Support Stations*** = sum('Banking Support'[Banking Stations])

Banking Support Stations

336

- ***Bonus Availability Stations*** = SUM('Bonus Availability'[Stations with Bonus])

Bonus Availability Stations

70

- ***Total Active Stations*** = sum('Active stations'[Open Stations Count])

Total Active Stations

2650

- ***Total Bike Stands*** = SUM(Bike_Fact[Bike Stands])

Total Bike Stands

57.2K

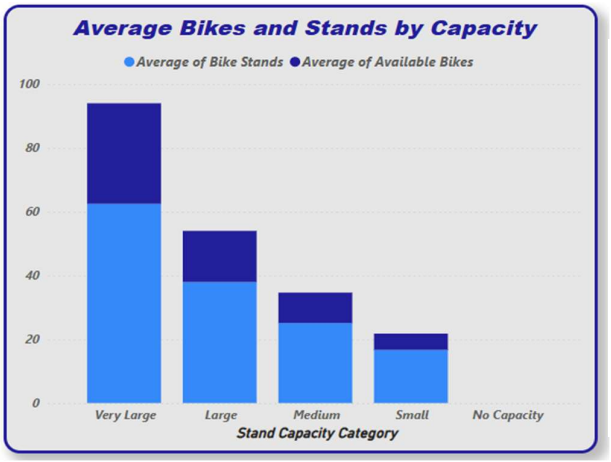
- ***Total Bike Stands Available*** = `SUM('Available Bike stands'[Bike Stand Count])`

- ***Total Bikes Available*** = `SUM('Available Bikes'[Total Available Bikes])`

Visualizations (Power BI)

Mutliple Visualizations:

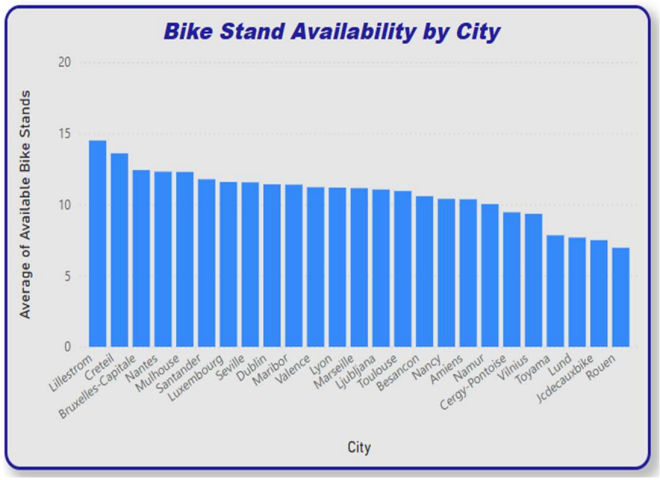
- ***Stacked Column Chart:***



Insights:

- **Very large stations** have the highest average number of bike stands and available bikes.
- **Bike availability decreases as station capacity becomes smaller.**
- **Small stations** provide limited bike availability due to fewer stands.
- **No capacity stations** have almost no bikes available.
- There is a positive relationship between station capacity and bike availability.

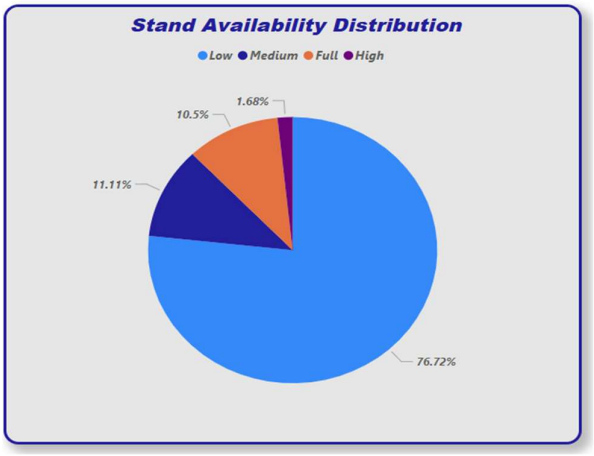
- ***Clustered Column Chart:***



Insights:

- **Lillestrom** has the highest average bike stand availability.
- Cities like **Créteil** and **Bruxelles-Capitale** also show relatively high availability.
- Most cities have **similar availability levels** (around 10–12 stands), indicating balanced distribution.
- **Rouen, Vélizy, and JCDecaux Lund** have the lowest average availability.
- Overall, the difference between cities is moderate rather than extreme.

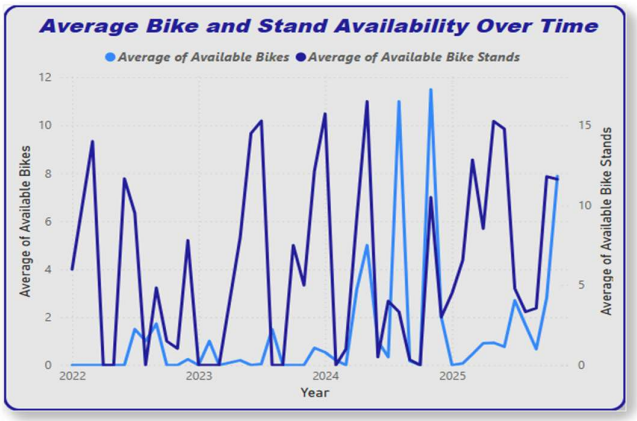
- ***Pie Chart:***



Insights:

- **Low availability** dominates with **76.72%** of stations.
- **Medium availability** accounts for **11.11%.**
- **Full stands** represent **10.5%**, indicating some stations reach full capacity.
- **High availability** is very rare at **1.68%**.
- Overall, most stations experience **low stand availability**, suggesting high usage.

- ***Area Chart:***



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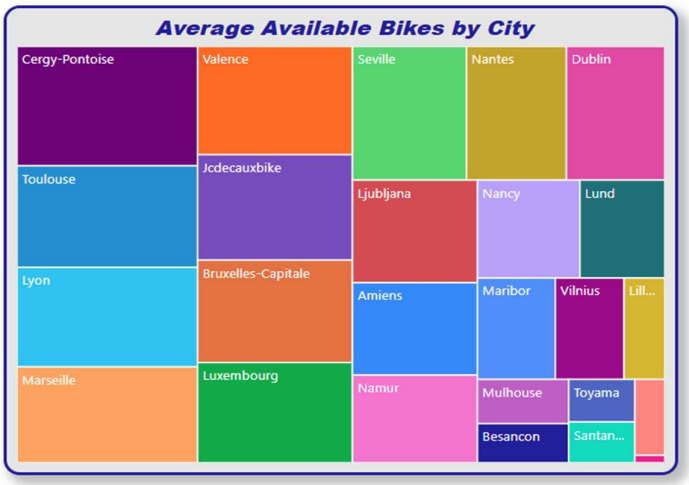
- **Full stands** represent **10.5%**, indicating some stations reach full capacity.
- **High availability** is very rare at **1.68%**.
- Overall, most stations experience **low stand availability**, suggesting high usage.

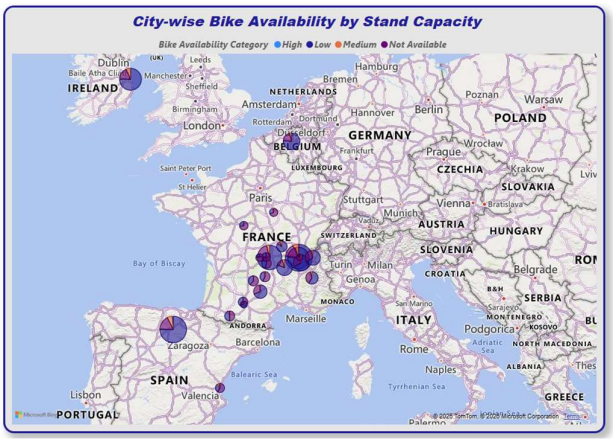
- ***Tree Map:***



Insights:

- Available bike stands **fluctuate** significantly over time.
- Available bikes remain relatively **low** compared to bike stands.
- A sharp **increase** in available bikes appears around **2024–2025**.
- The trend suggests **demand varies over time**, causing fluctuations in availability.
- **Bike stands are generally more available than bikes**, indicating frequent bike usage.

- ***Geographical Map:***



Insights:

- Most bike stations are concentrated in **France**, indicating **higher bike-sharing activity** there.
- Several cities show medium to high bike availability, but some locations have low or no bikes available.
- Bike availability varies significantly between cities, suggesting **uneven distribution**.
- A few cities **outside France** (e.g., Dublin and Zaragoza) also have **active bike stations**.
- The map highlights that bike availability **depends on stand capacity and city demand**.

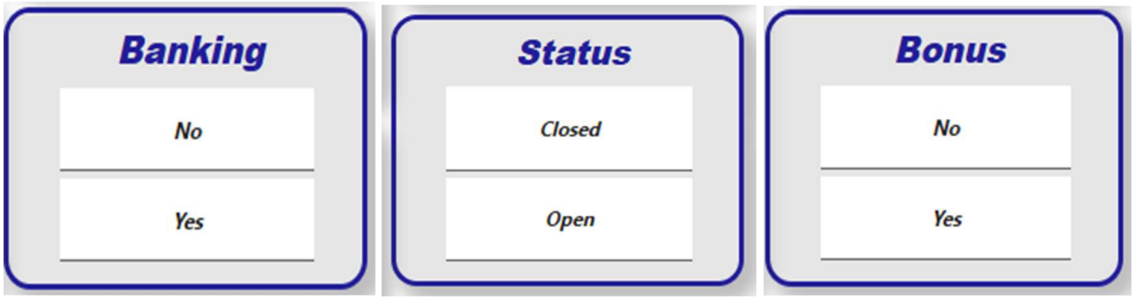
- ***Matrix:***

City	Sum of Bike Stands	Sum of Available Bikes	Sum of Available Bike Stands
Vilnius	653	117	346
Valence	5709	2116	3119
Toyama	379	32	196
Toulouse	8999	3583	4720
Seville	5499	1873	3133
Santander	326	22	212
Rouen	441	29	195
Nantes	2564	782	1588
Nancy	697	182	416
Namur	542	154	311
Mulhouse	754	75	504
Marseille	2611	1020	1450
Maribor	735	143	456
Lyon	9180	3654	5014
Luxembourg	3048	1043	1715
Lund	379	83	169
Ljubljana	1620	500	951
Lillestrom	143	15	116
Jcdecauxbike	198	89	90
Dublin	2236	693	1325
Creteil	178	1	136
Cergy-Pontoise	967	450	435
Bruxelles-Capitale	7392	2569	4435

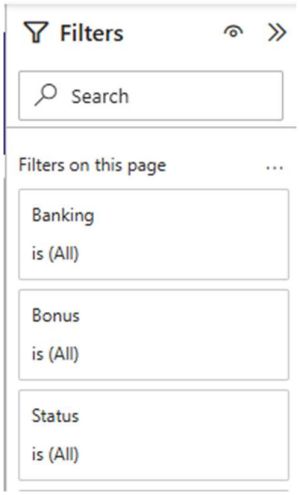
Insights:

- **Lyon and Toulouse** have the **highest number of bike stands and available bikes**, indicating large bike-sharing networks.
- **Bruxelles-Capitale** also shows **high bike and stand availability**.
- Cities like **Santander, Rouen, and Toyama** have very **low available bikes**, suggesting smaller systems.
- In most cities, **available bike stands are higher than available bikes**, meaning many bikes are currently in use.
- **Créteil** has almost **no bikes available**, indicating extremely high demand or low supply.

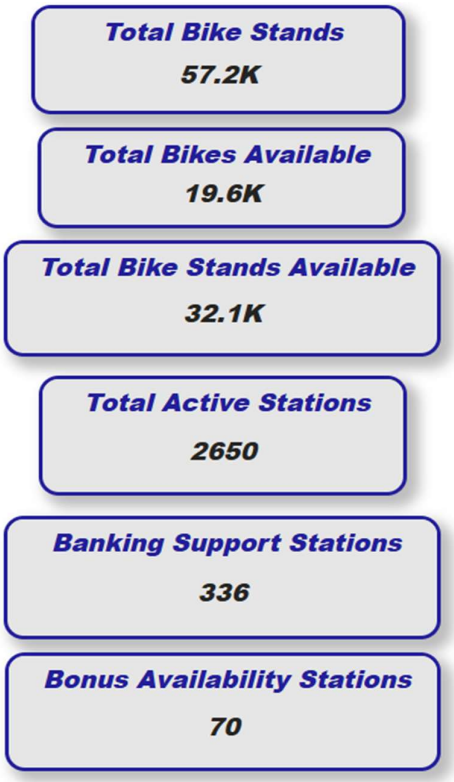
- ***Slicers:***



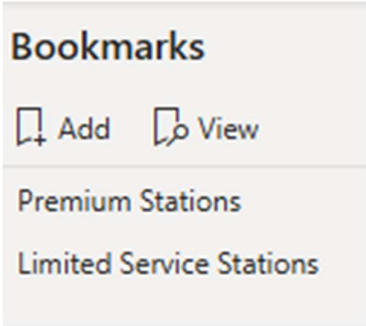
- ***Filters:***



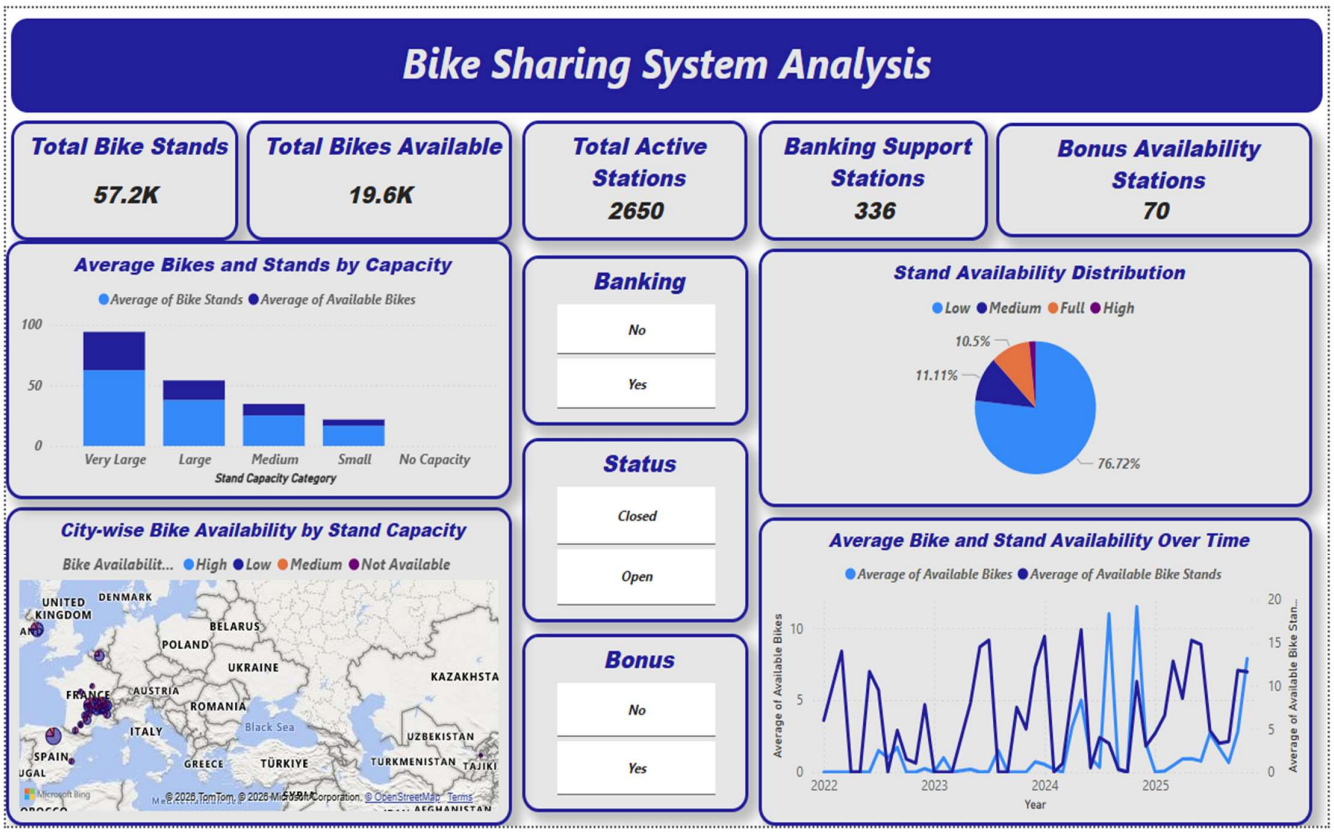
- ***Cards:***

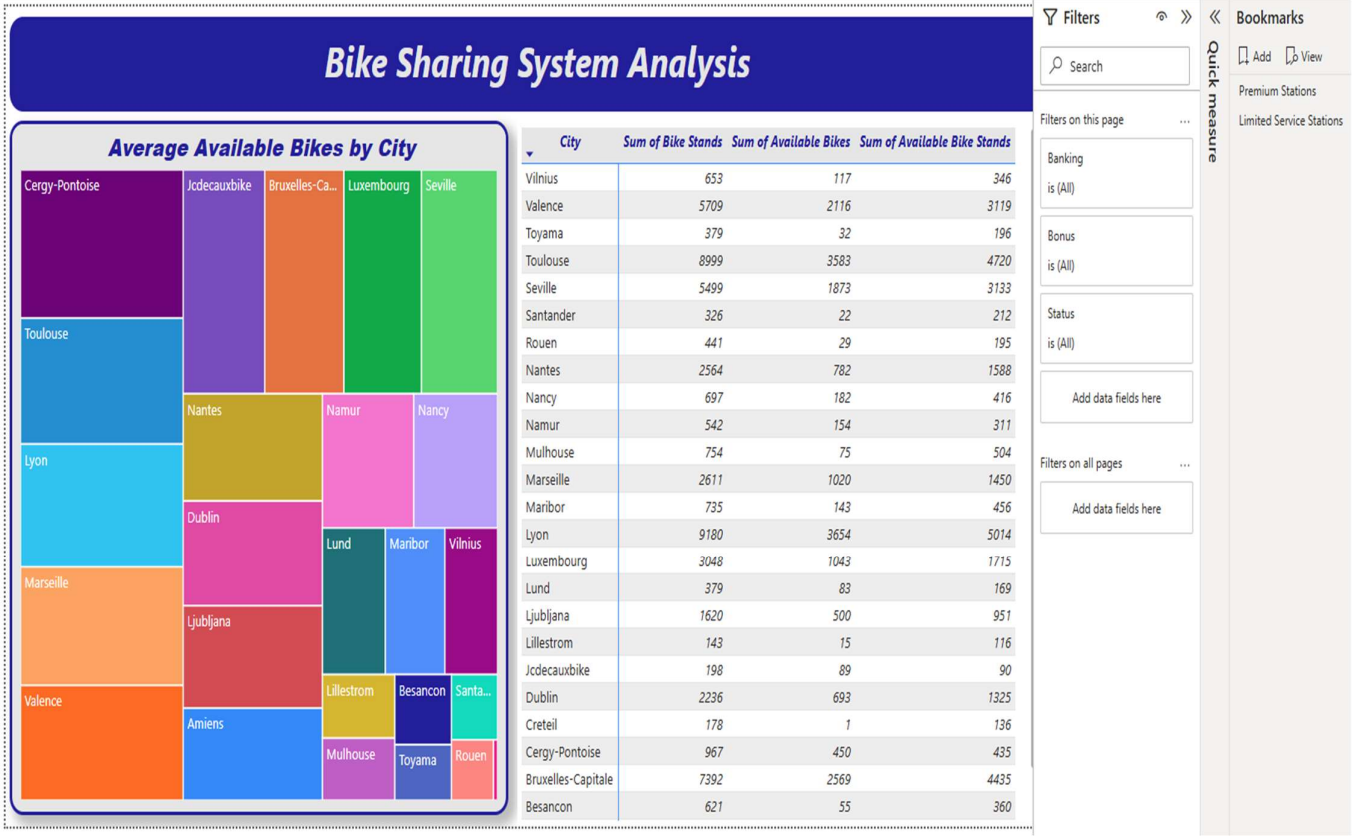


- ***Book mark/Buttons:***



Dashboards:





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